Chapter 6

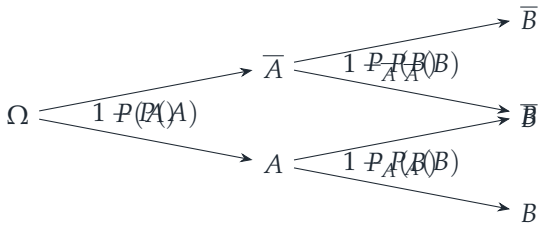
Conditional probabilities

Competences: model by tree or table, calculate, common difference / ratio on independence, interpret

6.1 Reference course

For two events A and B with $P(A) > 0$, the probability of B knowing A is

$$P_A(B) = P(B \mid A) = \frac{P(A \cap B)}{P(A)}.$$



Key ideas

In a weighted tree, the probabilities of the same node are as follows: 1. Probability along a path: $P(A \cap B) = P(A)P_A(B)$. Aggregate probabilities of incompatible paths leading to the desired event.

If $A_1, \ldots, A_n$ form a partition of the universe, so

$$P(B) = P(A_1)P_{A_1}(B) + \cdots + P(A_n)P_{A_n}(B).$$

Two events A and B non-zero probabilities are independent if $P(A \cap B) = P(A)P(B)$ or equivalent if $P_A(B) = P(B)$.

Frequent error

Do not confuse $P_A(B)$ and $P_B(A)$. The condition appears after the word "knowing" and is placed in index in the French notation.

Worked example

40% of customers buy online. Of these, 12% return their order; among others, Five percent turn it back. $P(R) = 0{,}40 \times 0{,}12 + 0{,}60 \times 0{,}05 = 0{,}078$. The overall rate of return is Thus, 7.8%.

Meeting 39 — Conditional probability

Conditional probabilities

09/03/2027 • B : 2 h – C : 1 h

Objectives of the session

· Define a conditional probability.

· Moving from strength to probability.

· Interpret.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 20 per cent of 140 .....
2. What is the value of the sum of branches from a node? .....
3. Calculate $0.7 \times 0{,}4$. .....
4. If $P(A) = 0.37$, calculate $P(\overline{A})$. .....
5. Simplify $\dfrac{18}{30}$. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, have the keywords reworded: define a conditional probability and pass counts with probabilities.

Research activity

In one survey, 180 out of 600 customers use a premium card and 72 of these 180 customers purchase Every week. $P(P)$, $P(P \cap H)$ and $P_P(H)$, then interpret.

Trace written

For $P(A) > 0$, $P_A(B) = \dfrac{P(A \cap B)}{P(A)}$. This is the probability of B in the population restricted to A.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. We know $P(A) = 0{,}35$, $P_A(B) = 0.6$ and $P_{\overline{A}}(B) = 0.8$. Build the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.
3. Calculate $P_B(A)$ and test independence.

Reference in the manual

Supplements: Calao, courses p. 132·137; exercises p. 138, n° 1 to 5.
session is identified below by its page and number; the reference also allows you to find
the figures and digital resources of the manual.

Manuel Calao · p. 138, exercise 5

One company buys electronic chips from three suppliers F_1, F_2 and F_3 : 30% in $F_1$20 per cent in F_2. Some fleas have defects: 5% of fleas coming from $F_1$3 per cent of those from F_2 and 10% of fleas F_3.
We pick a chip at random, or events. F_i : "The chip comes from the supplier i » and$D$: « The chip has a flaw".

1. Complete the manual's weighted shaft.
2. Determine the probability that the selected chip will have a defect.
3. Knowing that the chip has a defect, determine the probability that it comes from the supplier F_3.

Personal work

Resuming Exercise 2 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Session 40 — Weighted trees

Conditional probabilities

11/03/2027 • B : 1 h – C : 2 h

Objectives of the session

· Build and complete a tree.

· Read a conditional probability.

· Control the sums.

Fluency · 6 to 8 minutes, without calculator

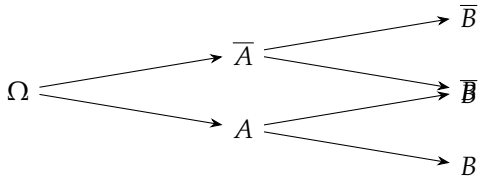
1. Write the multiplier coefficient of a 25 per cent increase . . . . . .
2. Calculate $0.8 \times 0{,}4$. . . . . .
3. If $P(A) = 0.37$, calculate $P(\overline{A})$. . . . . .
4. Simplify $\dfrac{18}{30}$. . . . . .
5. What is the value of the sum of branches from a node? . . . . . .

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: build and complete a tree and read a probability conditional.

Research activity

35% of orders are express. Of these, 4% are late; among standard orders, 4% are late.
9% are late. Build and complete the tree.



Trace written

In a tree, branches from a node have as a sum 1.
level are conditional.

Worked example

Retrieve activity data and write the solution in order: definition of objects,
property used, exact calculation or approach, then response phrase. The model response figure
in the teaching edition.

Training · essential

1. We know $P(A) = 0{,}40$, $P_A(B) = 0.2$ and $P_{\overline{A}}(B) = 0.4$. Build the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.

3. Calculate $P_B(A)$ and test independence.

Reference in the manual

Supplements: Calao, courses p. 133·137; exercises p. 140, n° 14 to 18.
session is identified below by its page and number; the reference also allows you to find the figures and digital resources of the manual.

Manuel Calao · p. 140, exercise 15

1. Complete the tree of the manual knowing that $P(A) = \frac{3}{5}$, $P_A(B) = \frac{7}{10}$, $P_A(D) = \frac{1}{5}$, $P_{\overline{A}}(B) = \frac{1}{3}$ and $P_{\overline{A}}(C) = \frac{2}{9}$.
2. Determine $P(A \cap B)$, $P(A \cap D)$ and $P(C)$.

Personal work

Resuming Exercise 3 with the following data:
320 and rate or common difference / ratio by 6. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Meeting 41 — Produces probabilities along a path

Conditional probabilities — 16/03/2027 • B : 2 h – C : 1 h

Objectives of the session

· Calculate an intersection.

· Combine paths.

· Translate an event.

Fluency · 6 to 8 minutes, without calculator

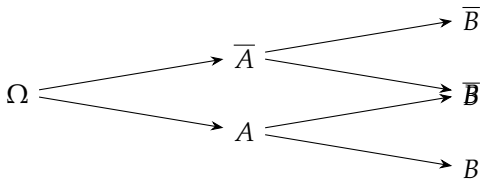
1. Reduce 3 $x + 9 - x + 2$. .....
2. If $P(A) = 0.37$, calculate $P(\overline{A})$. .....
3. Simplify $\dfrac{18}{30}$. .....
4. What is the value of the sum of branches from a node? .....
5. Calculate $0.9 \times 0{,}4$. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: calculate an intersection and combine paths.

Research activity

With the tree of session 40, calculate the probability of an express and late command, then the Total probability of delay. Interpret both results.



Trace written

We multiply along a path: $P(A \cap B) = P(A)P_A(B)$. Add incompatible paths that realize the event.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. We know $P(A) = 0{,}45$, $P_A(B) = 0.3$ and $P_{\overline{A}}(B) = 0.5$. Build the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.

3. Calculate $P_B(A)$ and test independence.

Reference in the manual

Supplements: Calao, p. 140, n° 14 to 18. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the digital resources of the manual.

Manuel Calao · p. 140, exercise 18

75% of the restaurant's guests are families, 17% of whom pay for their addition with 76% pay by credit card and the remainder pay in cash.
Of the other clients, only 5% use holiday vouchers, 80% use a credit card, and the rest of the liquid.
A table and its client(s) are randomly selected. F : "Customers are a family", C : "Customers pay in holiday vouchers", B : "Customers pay by credit card" and L : « Customers pay in cash."

1. Complete the manual's weighted shaft.
2. Describe with a sentence the event $\overline{F} \cap B$ then calculate the probability.
3. Calculate the probability that selected clients will pay with holiday vouchers.
4. The selected clients paid in holiday vouchers. Calculate the probability that these clients would be A family.

Personal work

Resuming Exercise 2 with the following data:
325 and the rate or common difference / ratio by 7. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Meeting 42 — Independence

Conditional probabilities

18/03/2027 • B : 1 h – C : 2 h

Objectives of the session

· Test independence.

· Compare probabilities.

· Justify a conclusion.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5 per cent of 60 .....
2. Simplify $\frac{18}{30}$. .....
3. What is the value of the sum of branches from a node? .....
4. Calculate $0.3 \times 0{,}4$. .....
5. If $P(A) = 0.37$, calculate $P(\overline{A})$. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: test independence and compare probabilities.

Research activity

In a population, $P(A) = 0{,}4$, $P(B) = 0{,}3$ and $P(A \cap B) = 0{,}12$. Test the independence of two ways. Then modify $P(A \cap B)$ in 0.15 and start again.

Trace written

A and B are independent if $P(A \cap B) = P(A)P(B)$, equivalently if $P_A(B) = P(B)$ where $P(A) > 0$.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. We know $P(A) = 0{,}20$, $P_A(B) = 0.4$ and $P_{\overline{A}}(B) = 0.6$. Build the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.
3. Calculate $P_B(A)$ and test independence.

Reference in the manual

Supplements: Calao, courses p. 134·137; exercises p. 144, n° 33 to 35.
session is identified below by its page and number; the reference also allows you to find

the figures and digital resources of the manual.

Manuel Calao · p. 139, exercise 13

At the exit of a factory, watches may show a defect (noted) D_1) on the bracelet and a default (noted) D_2) on the dial. A watch is called defective if it has at least one of the Two defects. A watch is taken at random in the production of a day.
Note A event: "The watch presents the defect D_1 » and B event: "The watch presents the defect D_2 The probabilities of these events are known: $P(A) = 0{,}02$ and $P(B) = 0{,}01$. On assumes that these two events are independent.

1. Determine the probability that the watch has both defects.
2. Determine the probability that the watch did not have any defects.

Personal work

Resuming Exercise 3 with the following data:
330 and the rate or common difference / ratio by 2. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Meeting 43 — Partition and total probabilities

Conditional probabilities

23/03/2027 • B : 2 h – C : 1 h

Objectives of the session

· Recognize a partition.

· Apply total probabilities.

· Organize the calculation.

Fluency · 6 to 8 minutes, without calculator

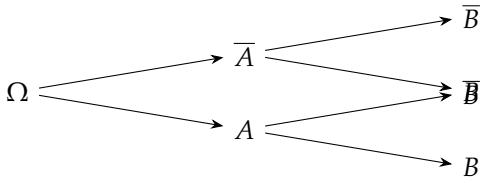
1. Write the multiplier coefficient of a 10 per cent increase . . . . . .
2. What is the value of the sum of branches from a node? . . . . . .
3. Calculate $0.4 \times 0{,}4$. . . . . .
4. If $P(A) = 0.37$, calculate $P(\overline{A})$. . . . . .
5. Simplify $\dfrac{18}{30}$. . . . . .

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: recognize a partition and apply probability total.

Research activity

One plant has three lines A, B, C producing 20%, 50% and 30% respectively.
Their default rate is 1%, 2%, 4%. Calculate the overall default rate.



Trace written

If (A_i) is a score, $P(B) = \sum_i P(A_i)P_{A_i}(B)$. Each term corresponds to a path leading to B.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. We know $P(A) = 0{,}25$, $P_A(B) = 0.5$ and $P_{\overline{A}}(B) = 0.7$. Build the tree.

2. Calculate $P(A \cap B)$ then $P(B)$.

3. Calculate $P_B(A)$ and test independence.

Reference in the manual

Supplements: Calao, p. 138, Nos. 1 to 5, and p. 142, Nos. 22 to 25.
is identified below by its page and number; the reference also allows to find the
figures and numerical resources of the manual.

Manuel Calao · p. 139, exercise 7

In the early 2000s, the Internet was not as popular and a study of the interest given to
The Internet on a sample of 2,000 people gave the following results.

Interested in the Internet?	Yes	No	Total
Under 30 years	560	...	700
30 to 60 years	...	...	...
Over 60 years	...	510	600
Total	800	...	2 000

1. Complete this table.

2. We pick a person we interviewed at random. A_1 : "The person is under 30 years of age. », A_2 : "The person is between 30 and 60 years of age", A_3 : "The person is over 60 years of age" and I : "The With the help of the table, complete the probability trees
of the manual.

Personal work

Resuming Exercise 2 with the following data:
335 and rate or common difference / ratio by 3. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and
indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Meeting 44 — Contextualized problems

Conditional probabilities

25/03/2027 • B : 1 h – C : 2 h

Objectives of the session

· Choose tree or table.

· Solve a contextualized problem.

· Interpret.

Fluency · 6 to 8 minutes, without calculator

1. Reduce 3 $x + 5 - x + 2$. .....
2. Calculate 0.5 $\times$ 0,4. .....
3. If $P(A) = 0.37$, calculate $P(\overline{A})$. .....
4. Simplify $\dfrac{18}{30}$. .....
5. What is the value of the sum of branches from a node? .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: choose tree or table and solve a problem Contextualized.

Research activity

Transform a client/subscription/resilience table into a tree, then respond to a question in each direction: probability of cancelling knowing the subscription; probability of having subscription knowing the termination.

Trace written

Choosing between tree and table: the tree highlights the order of conditioning; the table is well suited to staff and reverse readings.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. We know $P(A) = 0{,}30$, $P_A(B) = 0.6$ and $P_{\overline{A}}(B) = 0.8$. Build the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.
3. Calculate $P_B(A)$ and test independence.

Reference in the manual

Supplements: Calao, pp. 142, Nos. 22 to 25, and p. 144, Nos. 30 to 32.
session is identified below by its page and number; the reference also allows you to find
the figures and digital resources of the manual.

Manuel Calao · p. 142, exercise 24

A survey is carried out by a town hall to find out the habits of its administrators on the
waste treatment. 80% of respondents say they are sensitive to development
Of these, only 75% sort out their waste. 70% of respondents
declare that they are sorting their waste.
One of the people interviewed is chosen at random. D : "The person
is sensitive to sustainable development" and T : "The interviewee sorts the waste
».

1. Determine $P_D(T)$.
2. Determine $P_T(D)$.

Personal work

Resuming Exercise 3 with the following data:
340 and the rate or common difference / ratio by 4. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and
indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Meeting 45 — Mixed training

Conditional probabilities

30/03/2027 • B : 2 h – C : 1 h

Objectives of the session

· Calculate a reverse probability.

· Total chain and conditioning.

· Control the result.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 20 per cent of 120 .....

2. If $P(A) = 0.37$, calculate $P(\overline{A})$. .....

3. Simplify $\dfrac{18}{30}$. .....

4. What is the value of the sum of branches from a node? .....

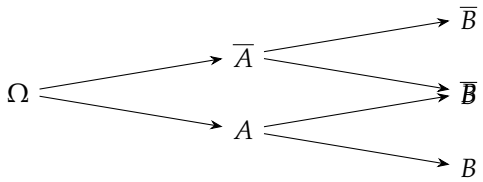
5. Calculate $0.6 \times 0{,}4$. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: calculate an inverse probability and sequence total and conditioning.

Research activity

In the situation of sitting 43, knowing that an article is defective, calculate the probability that from line C. Explain why this result is not 30%.



Trace written

For an inverse probability, first calculate the total probability of the conditioning event, then use the definition of conditional probability.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. We know $P(A) = 0{,}35$, $P_A(B) = 0.2$ and $P_{\overline{A}}(B) = 0.4$. Build the tree.

2. Calculate $P(A \cap B)$ then $P(B)$.

3. Calculate $P_B(A)$ and test independence.

Reference in the manual

Supplements: Calao, pp. 144-146. The exercise reproduced in this session is identified below. by its page and number; the reference also makes it possible to find figures and resources Numbers of the manual.

Manuel Calao · p. 143, exercise 26

In this exercise, the results will be given in the form of irreducible fractions.
to throw a balanced cubic dice then to pull a ball in a urn. If you get 6 at the throw of the dice, a ball is drawn in the A urn; if one obtains an odd number, one pulls in the B urn; in the other cases, fire in box C.
A urn contains 1 red ball and 5 green balls; B urn contains 2 red balls and 8 balls Green; C urn contains 3 red balls and 11 green balls.
The following events are considered: I : "Get an odd number"; S : "Get 6"; A : « Getting 2 or 4"; R : "The ball drawn is red"; V : "The ball drawn is green".

1. Represent a weighted tree corresponding to this situation.
2. Determine the probability of pulling a red ball.
3. Determine the probability of the ball coming from urn A knowing that it is red.

Personal work

Resuming Exercise 2 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Meeting 46 — Assessment and evaluation

Conditional probabilities

01/04/2027 • B : 1 h – C : 2 h

Objectives of the session

· Mobilize all methods.

· Write with notations.

· Remedy errors.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier coefficient of a 25 per cent increase . . . . . .
2. Simplify $\frac{18}{30}$. . . . . .
3. What is the value of the sum of branches from a node? . . . . . .
4. Calculate $0.7 \times 0{,}4$. . . . . .
5. If $P(A) = 0.37$, calculate $P(\overline{A})$. . . . . .

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: mobilize all methods and write with the ratings.

Research activity

Perform a four-post course: complete a tree; calculate an intersection; apply the total formula; test independence. Each group leaves a written justification to the group next.

Trace written

Always name events, complete all branches, write formula before calculation and interpret the final probability.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. We know $P(A) = 0{,}40$, $P_A(B) = 0.3$ and $P_{\overline{A}}(B) = 0.5$. Build the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.
3. Calculate $P_B(A)$ and test independence.

Reference in the manual

Supplements: Calao, p. 146·147. The exercise reproduced in this session is identified below. by its page and number; the reference also makes it possible to find figures and resources Numbers of the manual.

Manuel Calao · p. 143, exercise 29

In order to hire its managers, a company uses a recruitment firm.
a first selection on file. 40% of the files received are validated and forwarded to the company.
The selected candidates hold a first interview after which 70% of them are
The latter are invited to a final meeting with the Director of Resources
Human Resources, which recruits 25 per cent of the candidates met.
A candidate's file is randomly selected and the following events are considered: D : "The candidate is retained on file" ; E_1 : "The candidate is selected at the end of the first interview"; E_2 : "The candidate is recruited".

1. Complete the manual's weighted shaft. F Event: "The candidate is not recruited
   Demonstrate that the probability of the event F is equal to 0.93.

2. Five friends apply for executive employment in this company. Their files are being studied
   Determine the probability that at least one of the five friends will
   be recruited.

3. What is the minimum number of files that the recruitment firm needs to process in order to
   the probability of hiring at least one candidate is greater than 0.999?

Personal work

Resuming Exercise 3 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.